MILITARY SPECIFICATION

BATTERIES, DRY (SPECIAL PURPOSE)

1. SCOPE

1.1 Scope.- This specification covers special purpose batteries for use with highly specialized military equipment where a high degree of reliability is required. The battery is composed of electrochemical cells of the zinc-potassium hydroxide-mercuric oxide type.

1.2 Classification.

1.2.1 Type designation.- The type designation of mercury cell dry batteries shall be in the following form:

BA	1232/U
Component	Battery type number (1.2.1.1)

1.2.1.1 Battery type number.- The battery type number identifies the basic design of the battery (See 3.1) and consists of a four-digit number (in the 1001 thru 1999 series).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

FSC 6135

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SPECIFICATIONS

FEDERAL

FF-N-836	-Nut, Plain, Hexagon, Square, Cap and Welding Nut, Slotted, and Castellated, Hexagon.
FF-S-92	-Screw, Machine, Slotted or Cross-Recessed.
L-P-390	-Plastic, Molding Material, Polyethylene, Low and Medium Density.
L-P-513	-Plastic Sheet, Laminated, Thermosetting, Paper-Base, Phenolic-Resin.
QQ-B-613	-Brass, Leaded and Non-Leaded, Plate, Rolled Bar, Sheet and Strip.
QQ-B-626	-Brass, Leaded and Nonleaded, Rod, Shaped, Forgings, and Flat Products with Finished Edges, Bar and Strip.
QQ-B-750	-Bronze, Phosphor, Bar, Plate, Rod, Sheet, Strip, Flat Wire, and Structural and Special Shaped Sections.
QQ-C-502	-Copper Rods, and Shapes, and Flat Products with Finished Edges (Flat Wire, Strips and Bars).
QQ-C-530	-Copper-Beryllium Alloy Bars, Rods, and Wire.
QQ-C-533	-Copper-Beryllium Alloy Strip.
QQ-C-576	-Copper Flat Products with Slit, Slit and Edge-rolled, Sheared, Sawed, or Machined Edges, (Plate, Bar, Sheet and Strip).
QQ-N-290	-Nickel Plating (Electrodeposited).
QQ-P-416	-Plating, Cadmium (Electrodeposited).
QQ-S-571	-Solder: Lead Alloy, Tin Lead Alloy, and Tin Alloy; Flux Cored Ribbon and Wire, and Solid Form.
PPP-T-60	-Tape, Pressure-Sensitive Adhesive, Waterproof, for Packaging.

MILITARY

MIL-M-14	-Molding Plastics and Molded Plastic Parts, Thermosetting.
MIL-W-76	-Wire and Cable, Hook-up, Electrical Insulated.
MIL-B-857	-Bolts, Nuts, and Studs.
MIL-W-6858	-Welding, Resistance: Aluminum, Magnesium, Non-Hardening Steels or Alloys, Nickel Alloys, Heat-Resisting Alloys, and Titanium Alloys; Spot and Seam.
MIL-F-14256	-Flux, Soldering, liquid (Resin Base).
MIL-B-55521	-Battery, Dry Packaging and Packing.

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STANDARDS

FEDERAL

Fed. Std. No. 595	-Colors.

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-143	-Specifications and Standards, Order of Precedence for the Selection of.
MIL-STD-202	-Test Methods for Electronic and Electrical Component Parts.
MIL-STD-810	-Environmental Test Methods.

(Copies of specifications, standards, specification sheets, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Military specification sheets for individual battery types.- Detail requirements or exceptions applicable to individual types are specified on the military specification sheets of this specification. In the event of any conflict between requirements of this specification and the individual military specification sheet the latter shall govern.

3.2 Classification of requirements.- The requirements for the product are classified herein as follows:

Requirement	Paragraph
Selection of specifications and standards	3.3
Preproduction	3.4
Materials and components	3.5
Design and construction	3.6
Electrolyte penetration	3.7
Humidity	3.8
Capacity	3.9
Insulation Resistance	3.10
Vibration	3.11
Mechanical Shock	3.12
Labeling and marking	3.13
Workmanship	3.14

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3.3 Selection of specifications and standards.- Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Preproduction.- Batteries furnished under this specification shall be a product which has been tested, and passed the preproduction inspection specified herein.

3.5 Materials and components.- When a definite material or component is specified, it shall be in accordance with the applicable specification or requirement listed in table I. When deemed necessary by the Government, certification from the source of the material or component will be required. In the absence of certification from the source, a certificate of analysis or certified inspection data will be required. (See par. 4.4 and 4.4.1).

3.5.1 Metals.- All metals which do not enter into the basic electrochemical reaction of the cell shall resist or be treated to resist corrosion when subjected to any test or storage conditions specified herein.

3.5.1.1 Dissimilar metals.- When dissimilar metals are used in intimate contact with each other, protection against electrolysis and corrosion shall be provided.

3.6 Design and construction.- Batteries shall be of the design, construction, physical dimensions, weight, and polarity specified. (See 3.1).

3.6.1 Battery voltage.

3.6.1.1 Open-circuit voltage.- Unless otherwise specified, the open-circuit shall not exceed the specified nominal voltage by more than 15 percent. (See 3.1 and 4.7.5.1.)

3.6.1.2 Closed-circuit voltage.- The closed-circuit voltage shall be not less than the voltage specified. (See 3.1 and 4.7.5.2.)

3.6.2 Insulating, impregnating, potting and sealing compounds.- The insulating, impregnating, potting and sealing compounds shall exclude moisture from insulating material without impairing its electrical characteristics. When tested as specified in 4.7.1.1.1 the potting and sealing compounds shall not flow at high temperature (160°F) nor crack or draw away from the sides of a container at low temperature (0°F) sufficiently to impair electrical connections.

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TABLE I Materials and Components

Materials or components	Applicable specifications or requirements (See 3.5)	Methods of test (See 4.4 and 4.4.1)
Solder[1/]	QQ-S-571	
Soldering Flux[2/]	MIL-F-14256	
Metals	3.5.1	
Brass	QQ-B-613 or QQ-B-626	
Copper	QQ-C-502 or QQ-C-576	
Beryllium copper	QQ-C-530 or QQ-C-533	
Phosphor bronze	QQ-B-750	
Cadmium plating	QQ-P-416	
Nickel plating	QQ-N-290	
Wire	MIL-W-76	
Machine screws, studs and nuts	MIL-B-857, FF-N-836 or FF-S-92	
Plastic, laminated	L-P-513, type PBE	
Plastic, molded	MIL-M-14, type MFE	
Plastic, polyethylene	L-P-390, type I	
Tape	PPP-T-60	
Insulating, impregnating potting and sealing compounds	3.6.2	4.7.1 and 4.7.1.1.1
Filler or padding	3.6.3	4.7.1
Cell-block-container material	3.6.4 and 3.7	4.7.1 and 4.7.2
Intercell separation	3.6.5	4.7.1
Battery container	3.6.9 and 3.7	4.7.1 and 4.7.2

1/ For electrical connections, type Sn40 or higher tin content shall be used.
2/ If other fluxes are used, they shall not affect the performance of the battery or reduce its shelf life.

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3.6.3 Filler or padding.- Filler or padding shall be a cushioning electrically nonconducting material which maintains its insulating characteristics under adverse environmental conditions.

3.6.4 Cell-block container.- Cell-block container shall be an insulating material surrounding a group or a stack of individual cells.

3.6.5 Intercell separation.- A separator shall be placed between cells in series connected multicell batteries. The separator may be an absorbent material.

3.6.6 Intercell connections.- Intercell connections shall be spot welded in accordance with MIL-W-6858, class B. Connections between cell blocks and between cell block and terminal shall be so insulated or positioned as to avoid contact with other conducting material. When insulated wire is soldered to terminal lugs, it shall not be bared more than 3/32 inch from the lug nor shall it extend more than 3/32 inch beyond the lug.

3.6.7 Age of cells.- The minimum age of cells, from the time of their fabrication to the time of their presentation for acceptance inspection as batteries, shall be 5 days. The maximum age of cells from the time of their fabrication to the time of their shipping date shall be 90 days. Batteries shall be submitted for acceptance inspection not more than 30 days prior to the shipping date.

3.6.8 Terminals.- Terminals shall be provided as specified in individual specification sheets (3.1).

3.6.9 Battery container.- The battery container shall consist of either metallic or plastic material as specified in the individual specification sheets (3.1). The contents of the batteries shall be completely insulated from the container and sufficiently potted to minimize movement of individual cells. The color of exposed surfaces of battery containers shall match one of the following lusterless greens: 34079, 34086, 34087, 34096, 34102, 34127, and 34128 per Federal Standard No. 595.

3.7 Electrolyte penetration.- When applicable materials are tested as specified in 4.7.2, the milliammeter deflection shall be less than 0.225 ma during the entire test period.

3.8 Humidity.- After the batteries have been tested as specified in 4.7.3 they shall meet the capacity requirement of 3.1 and 3.9.

3.9 Capacity.- When the battery is tested for capacity as specified in 4.7.6, the discharge time prior to termination of test as required in 4.7.6.4 shall be not less than the minimum time specified. (See 3.1.)

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3.10 Insulation resistance.- The insulation resistance between any two terminals not electrically connected, and between all ungrounded terminals and the container of the battery, shall be not less than five megohms, when tested as specified in 4.7.7.

3.11 Vibration.- After the batteries have been tested as specified in 4.7.8 they shall meet the visual and mechanical, dimensional, closed circuit voltage and terminal requirements. (See 3.1, 3.6, 3.6.1.2 and 3.6.9.)

3.12 Mechanical shock.- After the batteries have been tested as specified in 4.7.9 they shall meet the visual and mechanical, dimensional, closed circuit voltage and terminal requirements. (See 3.1, 3.6, 3.6.1.2 and 3.6.9.)

3.13 Labeling and marking.- All labeling and marking shall be clear and legible throughout all of the tests specified herein. Labeling and marking shall be black. Battery containers may have the labeling and marking embossed or die depressed, in which case it may be the same color as the background.

3.13.1 Labels.- Each battery shall have label positioned as specified, in 3.1. If there is insufficient space to show all required information on one face of the battery, it shall be continued on another face. There shall be no information on the label other than the following:

BATTERY, DRY
Type designation
(Contract number) . . . .
(Code) . . . .
Manufacturer's name
(Trade name may also be used)
Manufacturer's plant

EXAMPLE:

BATTERY, DRY
BA-1232/U
DAAB05-69-C-1234
0369
John Doe Company
JODOCO
Batteryville, N.J.

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3.13.1.1 Code.- The code shown shall indicate the month and year of manufacture of the battery by means of a four-digit number in which the first two digits shall indicate the number of the month and the last two digits shall indicate the year. Months earlier than the tenth month shall be single digit preceded by "0."

Examples:

A battery manufactured in March 1969 will bear the code "0369."

A battery manufactured in November 1969 will bear the code "1169."

When a battery is completed during the last 3 working days of a month, or the first 3 working days of the subsequent month, the manufacturer is permitted to use either month as the date to be coded.

3.13.2 Terminal markings.- On batteries having socket-type terminals, all markings such as polarity, voltage, and the unit of battery (A,B,C, etc.) shall appear on the face of the battery bearing the socket. On other type terminals, the terminal markings may appear on the top or the side of the battery, or both. Markings shall indicate clearly the terminals to which they refer.

3.14 Workmanship.- Batteries shall be processed in such a manner as to be uniform in quality and shall be free from defects that will affect their life, serviceability, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities.- Test equipment and inspection facilities shall be of sufficient accuracy, quality and quantity to permit performance of the required inspection. The supplier shall establish calibration of inspection equipment to the satisfaction of the Government. Inspection equipment furnished by the Government, for performing certain inspection at supplier's plant, will be calibrated by the supplier and maintained by the Government. A written description of the inspection system shall be submitted to and be subject to review by the procuring activity.

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4.1.1.1 Instrument accuracy.

4.1.1.1.1 Voltmeters and ammeters.- All voltmeters and ammeters used in testing the batteries shall be accurate within 1 percent of the full-scale value. The voltmeter and ammeter ranges shall be such that all readings are taken on the upper half of the scale. The sensitivity of voltmeters shall be not less than 1,000 ohms per volt.

4.1.1.1.2 Resistor tolerances.- During all tests involving discharge through a resistance, such resistance shall be accurate within the following percentages:

	Percent
Up to an including 25,000 ohms	0.5
From above 25,000 ohms to and including 1 megohm	1.0
Above 1 megohm	5.0

In determining the resistance used as a test load, the resistance of all continuously operating voltmeters shall be considered as part of the specified load.

4.1.1.1.3 Timing.- Timing equipment shall be accurate within 0.5 percent.

4.2 Classification of inspection.- The examination and testing of batteries shall be classified as follows:

(a) Materials and components inspection. (See 4.4.)
(b) Preproduction inspection. (See 4.5.)
(c) Quality conformance inspection.
    (1) Inspection of product for delivery (See 4.6.1.)
    (2) Inspection of preparation for delivery. (See 4.6.2.)

4.3 Inspection conditions.- Except as otherwise specified herein, all examinations and tests shall be performed at a temperature of 80°±20°F. (26.7°±11.1°C.), ambient atmospheric pressure, and relative humidity.

4.4 Materials and components inspection.- Materials and components inspection shall consist of verification by certification from the source that the materials and components used in fabricating the batteries are in accordance with applicable requirements prior to such fabrication. In the absence of certification from the source a certificate of analysis or certified inspection data shall be required as proof of conformance to applicable requirements. Materials and components involved are listed in table I.

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4.4.1 Samples of materials and components.- For those items listed in table I for which the specification requirement does not reference a subsidiary specification: eight samples of materials or components, treated and processed as they would be in the finished batteries, shall be inspected.

4.5 Preproduction inspection.- After award of contract preproduction inspection shall be performed by the supplier as specified in 4.5.1 through 4.5.1.3.

4.5.1 Sample batteries.- The supplier at no additional cost to the Government shall make 61 batteries constituting a preproduction inspection lot.

4.5.1.1 Inspection Routine.- Preproduction inspection shall consist of Groups I, II, III & IV per table II. The applicable tests for Group I will be specified in the appropriate individual specification sheet (3.1). Group II and III (Conditions A, B, C, & D) shall apply to all sixty (60) batteries being tested. Group IV shall consist of one sample battery, untested, to remain at the suppliers plant to be available as a standard for comparative purposes.

TABLE II - Preproduction Inspection

Group	No. of Batteries	Examination & Test	Req. Para.	Method of Test Para.
I	60	Vibration test	3.11	4.7.8
	60	Mechanical shock test I	3.12	4.7.9.1
	60	Mechanical shock test II	3.12	4.7.9.2
	60	Mechanical shock test III	3.12	4.7.9.3
II	60	Insulation resistance test	3.10	4.7.7
	60	Visual-mechanical examination	3.6 & 3.14	4.7 & 4.7.4
	60	Battery voltage	3.6.1	4.7.5
III	60	Capacity tests	3.9	4.7.6
IIIA	15	CycleLoad I or II	3.9	4.7.6
IIIB	15	Cont. Disch. @ 20°F	3.9	4.7.6
IIIC	15	Cont. Disch. @ 70°F after 1 month storage at 160°F	3.9	4.7.6
IIID	15	Cont. Disch. at 70°F after humidity test	3.8 & 3.9	4.7.3 & 4.7.6
IV	1	Untested - reference sample	---	---

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4.5.1.2 Preproduction Inspection Report.- Contractor shall furnish three (3) legible copies of complete test data for all tests performed by contractor on the preproduction inspection lot samples for approval within ten (10) calendar days of completion of testing. Reports shall be sent at contractor's expense to the procuring activity.

4.5.1.3 Failure.- If one or more sample batteries fail to meet any of the preproduction examinations and tests, the supplier shall immediately make necessary remedial changes. The supplier at no additional cost to the Government shall be required to submit additional preproduction samples for reinspection. A description of the corrective action taken or to be taken shall be included in the preproduction inspection report (see 4.5.1.2). Official approval to begin production will be given upon successful completion of preproduction inspection.

4.5.2 Start of production.- Any production prior to materials and components inspection and approval of the preproduction samples shall be at the supplier's risk (see 4.4 and 4.5).

4.6 Quality conformance inspection.

4.6.1 Inspection of product for delivery.- The contractor shall perform the inspection specified in 4.4 and 4.6.1.2 through 4.6.1.4.1. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government---at its discretion---may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.3). Test equipment for Government verification inspection shall be made available by the contractor.

4.6.1.1 Inspection Lot.- An inspection lot shall be defined as the quantity of batteries of any one type, of any one code, and produced at any one place of manufacture on any one contract, submitted at one time to quality conformance inspection.

4.6.1.2 Group A Inspection.- Each unit of each lot of batteries shall be inspected for conformance to all the inspection and test requirements of table III. Each lot will then be subject to verification, utilizing the procedures of Military Standard MIL-STD-105, and AQL's indicated in table III.

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TABLE III.- Group A inspection

Examination and Test	Requirement Paragraph	Method of Test Para.	AQL		Inspection level
			Major	Minor	
Visual & Mechanical Examination (External)	3.6 and 3.14	4.7.1 & 4.7.4	0.25%	1.0%	II
Battery Voltage	3.6.1 thru 3.6.1.2	4.7.5	0.25%	---	II
Insulation Resistance	3.10	4.7.7	0.25%	---	II

4.6.1.3 Group B Inspection.- This inspection shall consist of the continuous discharge at 20°F test (see 3.9 and 4.7.6). Sampling shall conform to procedures of Standard MIL-STD-105 with a general inspection level of I and an AQL of 2.5%. Group B inspection shall be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. Sample batteries subjected to group B inspection shall be in addition to quantity furnished under contract.

4.6.1.4 Group C Inspection.- This inspection shall consist of the tests specified in table IV and shall be performed on sample units that have been subjected to and met group A inspection. Sample units shall be selected in accordance with table V. Shipments shall not be withheld pending the results of group C inspection. (See 4.6.1.4.1.) Sample units submitted to group C inspection shall not be furnished on the contract.

TABLE IV.- Group C inspection

Examination and Test	Requirement Paragraph	Method of Test Para
Vibration test	3.11	4.7.8
Mechanical shock test I, II and III, as applicable	3.12	4.7.9 thru 4.7.9.3
Capacity		
*(a) Cyclic load I or II	3.9	4.7.6
*(b) Cont. Disch. at 70°F after 1 month storage	3.9	4.7.6
*(c) Cont. Disch. at 70°F after humidity test	3.9	4.7.6

*1/3 of sample size will be submitted to capacity test (a), 1/3 to (b) and 1/3 to (c).

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TABLE V.- Group C Sampling

Inspection Lot Size	Sample Size	Acceptance No.
7-150	6	0
151-500	21	1
501-1200	33	2
1201-3200	51	3
3201 and over	81	5

4.6.1.4.1 Non-Compliance.- When the total number of sample batteries with capacity values below the minimum requirements specified (see 3.1) for tests (a), (b), and (c) of table IV is greater than the applicable acceptance number stipulated in table V, release of batteries will be withheld pending results of corrective action. The supplier shall immediately report in writing the extent of the failure and characteristics affected. The supplier shall immediately investigate the cause of failure and further report the results of the investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and, (ii) all units of product which were manufactured under the same conditions and which are considered to be subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken to the satisfaction of the purchasing activity, additional sample batteries shall be subjected to group C inspection (all inspections, or the inspections which the sample failed, at the option of the purchasing activity) and groups A and B inspection may be reinstituted; however final acceptance and shipment will be withheld until the group C reinspection results have shown that corrective action was effective. At the option of the procuring activity, the supplier may be required to replace or correct, at his expense, all batteries considered subject to the same failure that have been shipped between the previous group C inspection and the group C inspection where failure occurred.

4.6.2 Inspection of preparation for delivery.- Inspection of preparation for delivery shall be accomplished in accordance with the quality assurance provisions of Military Specification MIL-B-55521.

4.7 Methods of examination and tests.

4.7.1 Visual and mechanical examination.- Batteries shall be examined to determine compliance with all applicable requirements and characteristics listed in table VI. When internal examination is necessary it shall be performed either during fabrication of the battery, or by dismantling the finished battery.

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TABLE VI.- Visual and Mechanical Examination

Requirement	Reference Paragraph
External	
Design and construction	3.6
Terminals	3.6.8
Battery containers	3.6.9
Dimensions and Weight	3.1
Internal	
Dissimilar metals	3.5.1.1
Filler or padding	3.6.3
Cell-block container	3.6.4
Intercell separation	3.6.5
Intercell connections	3.6.6
Age of cells	3.6.7
Battery container	3.6.9
Insulating, impregnating, potting and sealing and compounds	3.6.2

4.7.1.1 Potting.- The manufacturer shall determine the minimum weight of an adequately potted battery for each production lot, and shall not present for acceptance any batteries failing to meet this minimum weight requirement. This shall be done by determining the minimum weight of a battery filled with a sufficient quantity of potting material to withstand the vibration and mechanical-shock tests specified (see 3.1).

4.7.1.1.1 Flow or shrinking.- Potting and sealing compounds, when used, shall be placed in a container, approximately 3 inches wide by 6 inches long by 3/4 inch high, to within $\frac{1}{4}$ inch of the top. The temperature of the compound within the container shall be raised to $160^{\circ}\pm5^{\circ}$F. ($71.1^{\circ}\pm2.8^{\circ}$C.) and the container shall be held in an inverted position for 24 hours. Then the temperature of the compound shall be lowered to $0^{\circ}\pm5^{\circ}$F. ($-17.8.^{\circ}\pm2.80^{\circ}$C). The flow or cracking or shrinking of the compound from the sides of the container shall be noted. If flow is noted at $160^{\circ}\pm5^{\circ}$F. ($71.1^{\circ}\pm2.8^{\circ}$C.) five batteries of type involved shall be exposed to $160^{\circ}\pm5^{\circ}$F. ($71.1^{\circ}\pm2.8^{\circ}$C.) storage for 24 hours with terminals resting in lowest possible position. At the end of this exposure period batteries shall be examined to determine that there is no impairment of electrical contact (see 3.6.2).

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4.7.2 Electrolyte penetration.- Applicable specimens of materials and components shall be placed on a metal plate, and a cylindrical plastic tube 2 inches high with an internal diameter of $1\frac{1}{2}$ inches shall be clamped over the material. The tube shall then be filled to a depth of 1 inch with an aqueous solution of the appropriate electrolyte (potassium hydroxide 31 percent by weight). A metal electrode shall be inserted into this solution to a depth of approximately $\frac{1}{2}$ inch. A direct-current potential of 22.5 volts shall be applied between this electrode and the metal plate with a milliammeter of proper range in the circuit. The test shall continue for a period of 72 hours, but may be terminated if failure occurs sooner.

4.7.3 Humidity test.- Unless otherwise specified, the humidity test shall be performed. Batteries shall be tested in accordance with Standard MIL-STD-202, Method 103B, Test condition C. After drying period, the batteries shall be subjected to capacity discharge test at the 20 hour rate at 70°F. (4.7.6).

4.7.4 Dimensions and weight.- Batteries shall be examined by gaging or measuring and by weighing to determine conformance.

4.7.4.1 Dimensions.- All dimensions shall include any coating which may be used, and shall remain within the specified tolerances throughout the required tests. When box gages are used, batteries, loaded with the following weights, shall pass freely through the applicable gage openings:

a. Batteries weighing less than 5 pounds - loading weight of 5 pounds.

b. Batteries weighing 5 pounds or more - loading weight equal to the weight of the battery.

The inside dimensions of the box gage shall be the specified maximum outside dimensions of the battery.

4.7.5 Battery Voltage.

4.7.5.1 Open-circuit voltage.- A direct current voltmeter of proper range and sensitivity shall be used to measure the open-circuit voltage.

4.7.5.2 Closed-circuit voltage.- A direct current voltmeter of proper range and sensitivity shall be used to measure the closed circuit voltage utilizing resistance specified (see 3.1).

4.7.6 Capacity.- Sample batteries selected for capacity test shall be stored, if applicable, and discharged as specified in 4.7.6.2 through 4.7.6.4.

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4.7.6.1 Ambient storage and discharge conditions.- The ambient conditions specified in table VII shall prevail during storage and discharge periods. Normal conditions shall be maintained insofar as possible. Deviations from normal conditions are permitted, provided that: (1) the extreme conditions specified in table VII do not exist for more than five percent (cumulative) of the specified storage or discharge periods; and (2) that at no time are the extreme conditions exceeded.

4.7.6.2 Storage.- Sample batteries (unpackaged) shall be stored at applicable storage conditions for the specified period (see 3.1). The storage period shall be considered to have started from the date the batteries are actually placed in storage. At the conclusion of the storage period the outside of the battery container shall be examined for corrosion (see 3.6.9).

4.7.6.3 Stabilization preceding discharge.- Following storage, and conditioning, when applicable, the batteries shall be further stored for 48 hours at ambient discharge conditions specified in table VII.

4.7.6.4 Discharge.- Following stabilization the batteries shall be discharged at the ambient discharge conditions specified in table VII. The discharge shall be terminated when any one of the following conditions occur:

(a) The battery voltage or the voltage of any one unit falls below the specified test-end voltage. (For batteries requiring discharge alternately through two resistances, the voltage shall be read during the final minute of the heavier-load period.)

(b) The battery dimensions exceed the maximum specified.(See 3.1.)

(c) Corrosion becomes apparent on the exterior surface of the battery.

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TABLE VII.- Ambient Storage and Discharge Conditions

Kind of storage or discharge	Normal conditions		Extreme conditions	
	Temperature	Relative Humidity	Temperature	Relative Humidity
160° storage	160° +3°F -7°F	NA	140°F (60.0°C) through 153°F (67.2°C) and 163°F (72.8°C) through 165°F (73.9°C)	NA
70°F discharges Cyclic load I 1/ Cyclic load II 2/	70° +2°F (21.1° + 1.1°C.)	50 + 15	65°F (18.3°C.) through 68°F. (20°C) and 72°F. (22.2°C.) through 80°F. (26.7°C.)	25 through 35 65 through 75
20°F discharge Cont. Discharge	20° + 3°F (-6.67° + 1.67°C.)	NA	NA	NA

1/ Cyclic load I. Each battery will receive 360 pulses of 60 millisecond duration for the first hour of operation and 36 pulses of 60 millisecond duration for each hour of the next 23 hours. The entire sequence shall be repeated 45 times. The resistance load for the pulse will be specified in the individual specification sheet (see 3.1).
2/ Cyclic load II. Each battery will receive a 6 second pulse every 10 minutes. The pulse load and duration of test will be specified in the individual specification sheet (see 3.1).

4.7.7 Insulation resistance.- Insulation-resistance test shall be performed, except as otherwise specified. (See 3.1.) Batteries shall be stored for a period of 48 hours at 70°+5°F. (21.1°+2.8°C.) and a relative humidity of 50 + 15 percent. After storage and while at these conditions, the insulation resistance shall be measured by applying a direct-current potential of 500 + 20 volts between any two terminals not electrically connected and between all ungrounded terminals and the container of the battery. The insulation resistance of batteries having a nonmetallic container shall be measured by the use of a 1 inch-square copper plate making physical contact with the container. The plate shall be placed with the broad surface against any area of any surface of the container other than that on which the battery terminals are located. (See 3.10.)

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4.7.8 Vibration.- Except as otherwise specified the vibration test shall be performed. Each battery shall be rigidly clamped to the platform of a vibration machine in a manner approximating as closely as practicable the manner in which the batteries are clamped when in use. (See 3.1.) A simple harmonic motion having an amplitude of 0.03 inch (0.06 inch maximum total excursion) shall be applied. The frequency shall be varied at the rate of 1 cycle per second per minute between the limits of 10 and 55 cycles per second. The entire range of frequencies and return shall be traversed in 95 ± 5 minutes for each mounting position (direction of vibration) of the battery. The batteries shall be vibrated in three mutually perpendicular directions, for approximately equal periods. One of the directions of vibration shall be perpendicular to the terminal face of the battery. Open-circuit voltage shall be observed for 30 seconds during the last quarter of each vibration period.

4.7.9 Mechanical shock.- When more than one test is specified, the tests shall be performed in numerical order of the numbers designating the tests. Each battery shall be secured to the testing machine by means of a rigid mount which will support all mounting surfaces of the battery. (See 3.1 and 3.12.)

4.7.9.1 Mechanical-shock test I.- Each battery shall be subjected to a total of three shocks of equal magnitude. The shocks shall be applied in each of three mutually perpendicular directions. Each shock shall be applied in a direction normal to a face of the battery. For each shock, the battery shall be accelerated in such a manner that during the first 3 milliseconds the minimum average acceleration is 75 gravity units (g). The peak acceleration shall be between 125 to 175 G.

4.7.9.2 Mechanical-shock test II.- This test shall be conducted only when specified in battery specification sheet. Batteries exclusive of packaging or other protective means such as battery compartments shall be tested in accordance with Military Standard MIL-STD-810, Method 516, Procedure II (Transit drop test).

4.7.9.3 Mechanical-shock test III.- This test shall be conducted only when specified in the battery specification sheet. The battery shall be subjected to single shock pulse of 1000 g's 5 millisecond duration sine wave shape along each axis. The terminal face shall be opposite to the impact base.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking.- Preservation, packaging, packing and marking shall be in accordance with Military Specification MIL-B-55521. Preservation and packaging shall be level A or C with each unit package containing one battery, and packing shall be level A, B or C, as specified in contract or order (see 6.1(d)).

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6. NOTES

6.1 Ordering data.- Contractual documents should specify the following:

(a) Title, number, and date of this specification.
(b) Complete battery type designation and the title, number and date of the applicable specification sheet (see 1.2.1 and 3.1).
(c) Date for notice of availability for shipment.
(d) Applicable levels of preservation, packaging and packing.

6.1.1 Indirect shipments.- The packaging, packing, and marking specified in section 5 apply only to direct purchases by or direct shipment to the Government and are not intended to apply to contracts or orders between the supplier and prime contractor.

6.2 Awarding of contract.- Contracts will be awarded only to suppliers who guarantee to meet the requirements of this specification. No combining of performance requirements should be undertaken, Bids that offer to guarantee higher capacities will receive no special consideration in awarding a contract. Contracts will be awarded to the lowest bidder on a cost-per-unit battery basis provided that all performance requirements are guaranteed. Failure on a prior contract, of a manufacturer's particular battery type to meet any of the performance requirements of this specification, will be adequate cause for rejection of bids on that particular type until the supplier submits certified data providing that--

(a) Action has been taken to eliminate the cause of all failures; and
(b) The battery meets all the performance requirements of this specification.

6.3 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

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